

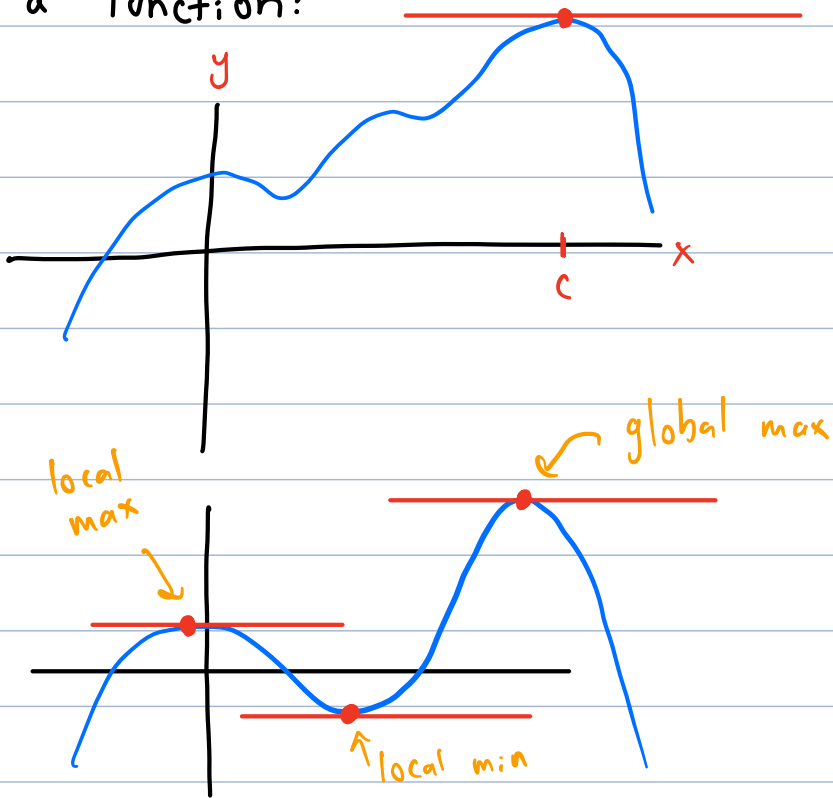
Calc I Lecture 16: Max and min

Today we discuss a very useful application of derivatives.
finding minimum and maximum values of functions.

Often we want to find the best solution to some problem.

This can be expressed by maximizing or minimizing some quantity
(think: cheapest, quickest, biggest, most profitable, ...)
min max

We can get some insight into this by looking at the graph of a function:



observation:

$$f'(c) = 0$$

where f attains its
max value at c

preliminary observation:

- if a differentiable function f attains a maximum value at some point a , then $f'(a) = 0$
- similarly if f attains a minimum value at a , then $f'(a) = 0$

(but careful: the other direction is not true)

$f'(a) = 0$ does not imply f attains min or max value at a

Terminology:

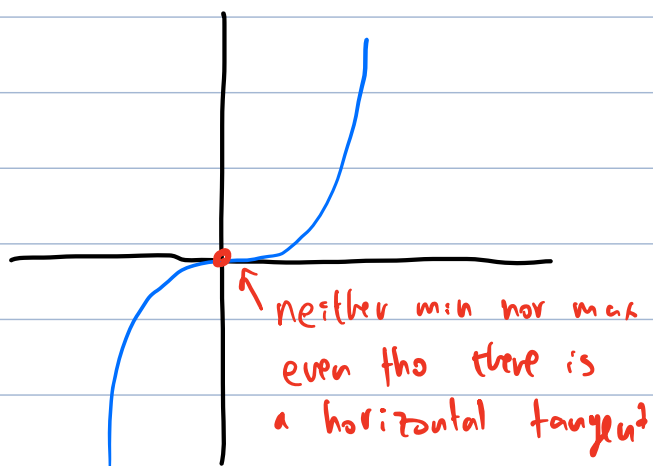
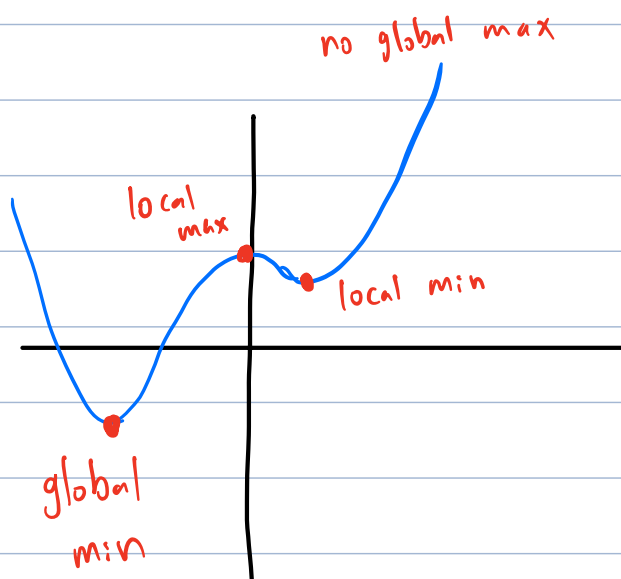
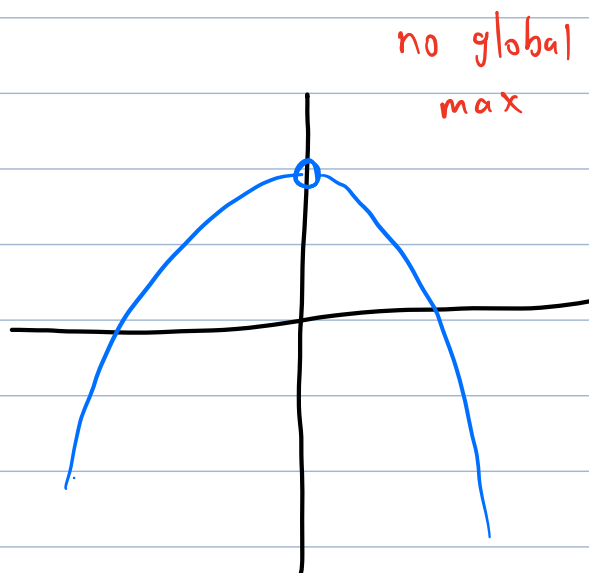
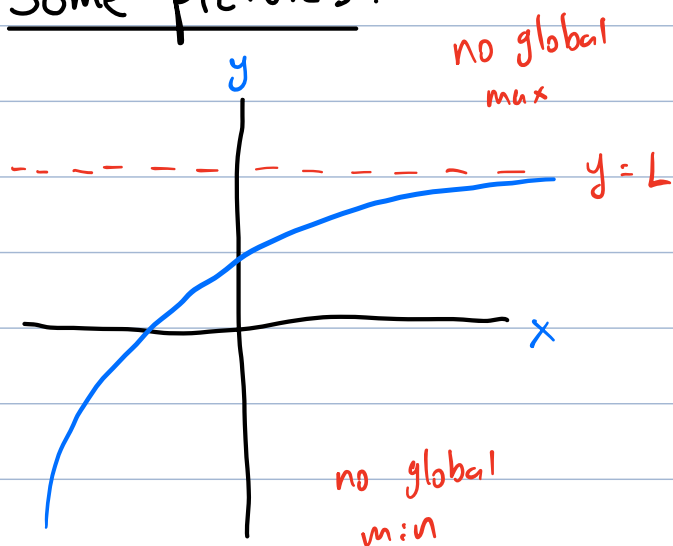
- A function f has an absolute maximum (aka global maximum) at $x=c$ if $f(c) \geq f(x)$ for all x .
The number $f(c)$ is called the maximum value of the function
y-value
- Similarly, f has an absolute minimum (aka global minimum) at $x=c$ if $f(c) \leq f(x)$ for all x .
 $f(c)$ is the minimum value of the function
- The maximum and minimum values of f are called the extreme values of f .

Next, we consider the local behavior of functions:

- A function f has a local maximum (aka relative maximum) at c if $f(c) \geq f(x)$ for x near c . (if we zoom in close enough to c)
^ both to the right and to the left
- Similarly, f has a local minimum at c if $f(c) \leq f(x)$ for x near c

Technicality: a global max is also a local max
a global min is also a local min.

Some pictures:



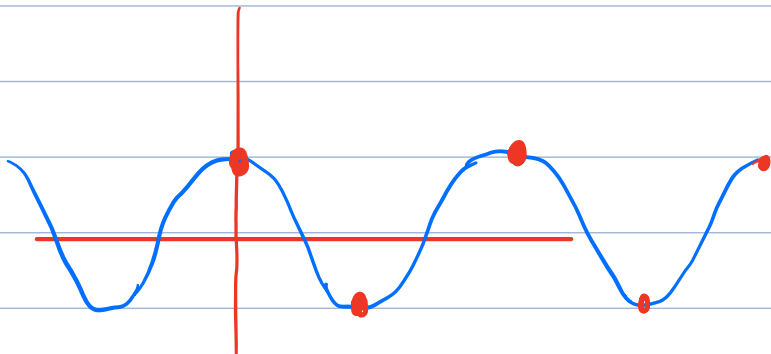
Example: Find local and global max and min for the following functions, if they exist (no derivatives needed)

a) $y = \cos(x)$

max value: 1

min value: -1

there are no other local min/max.

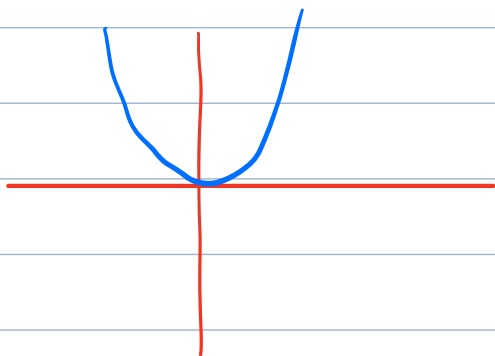


b) $y = x^2$

no global max

global min value: $y = 0$

attained at $x = 0$

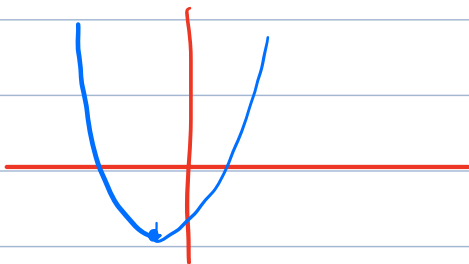


b2) $y = (x+1)^2 - 5$

no global max

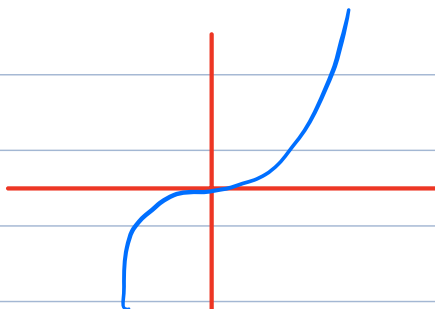
global min value: -5

attained at $x = -1$

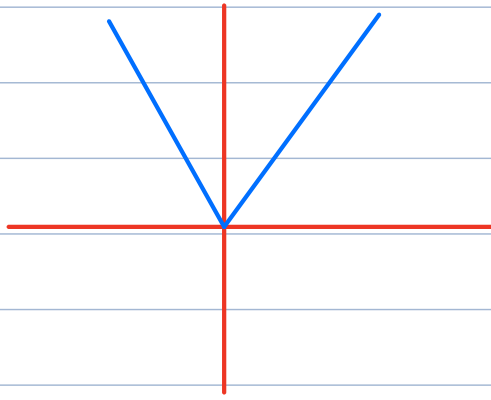


c) $y = x^3$

no local max
or min



d) $y = |x|$



no global max

global min value = 0
attained at $x = 0$

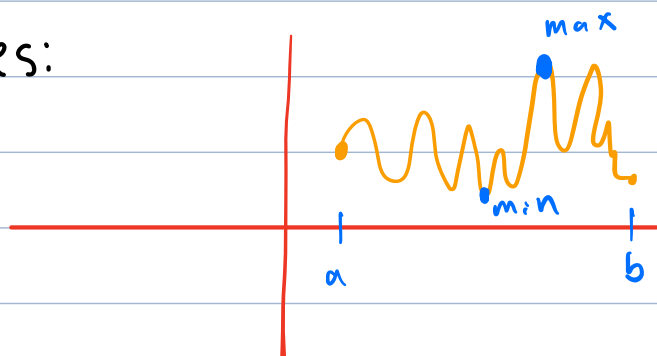
observation: some functions do not have any extreme values

But there are some conditions that guarantee they exist:

The extreme value Theorem: if f is continuous on a closed interval $[a, b]$, then

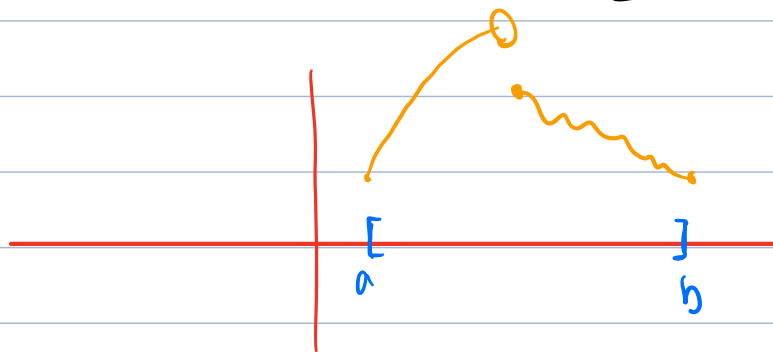
f attains an absolute maximum value } over the interval $[a, b]$.
and an absolute minimum value
at some points in the interval $[a, b]$.

Pictures:



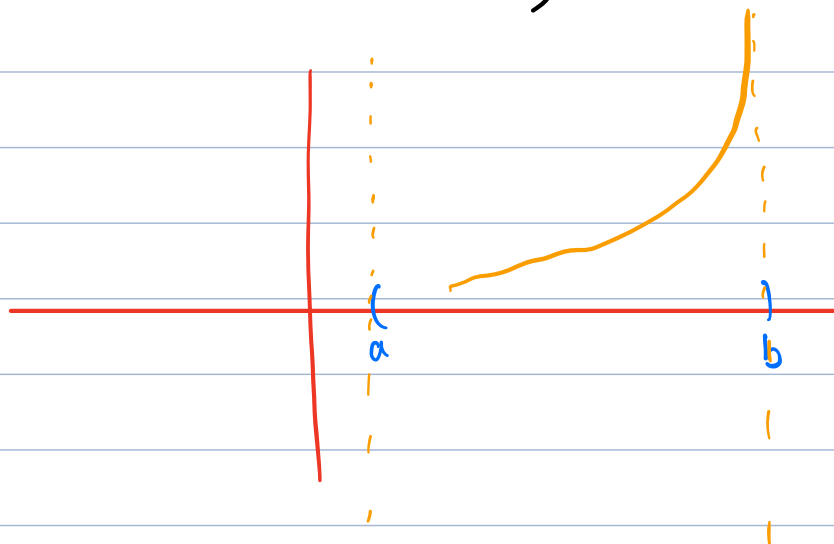
Pictures where the conditions fail:

- if the function is not continuous: no longer true



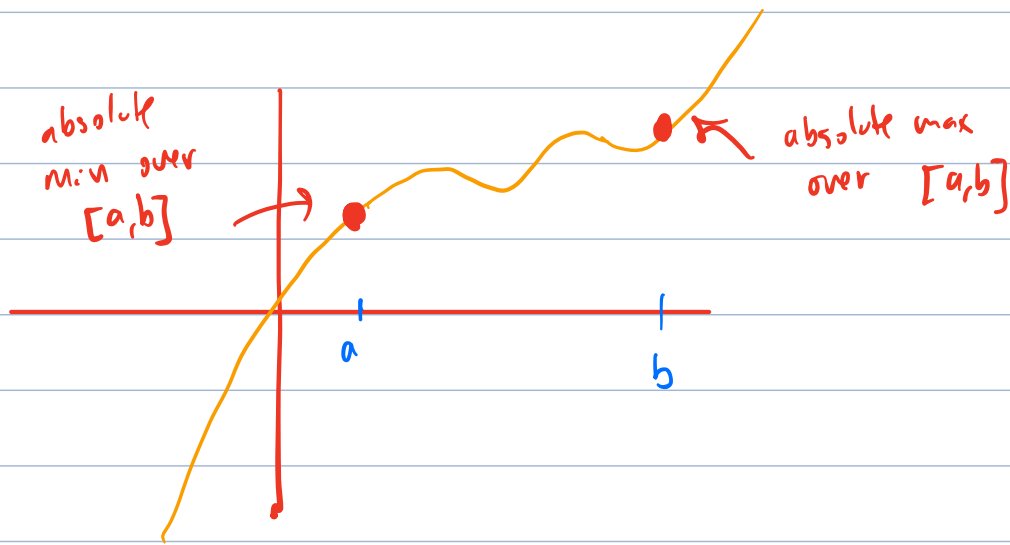
no global max
(you never reach it)

- if the function is only defined over an open interval (a, b)



no global min
or max

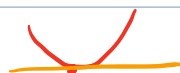
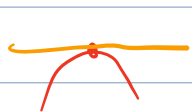
Note: the absolute min and max could be attained at $x=a$ or $x=b$ potentially, even if it's not a local min/max



Finally we return to derivatives:

Fermat's Theorem:

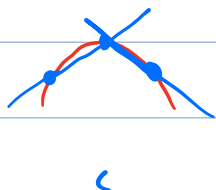
if f has a local max or min at $\underline{x=c}$, and if $\underline{f'(c)}$ exists, then $f'(c)=0$.



Proof sketch:

Remember: tangent line is a limit of sec lines.

If f has local max at c



secant lines from right have neg. slope

secant lines from left have pos. slope

$\Rightarrow$ limit of slopes of secant lines

has to be ≥ 0 and ≤ 0

$\Rightarrow$ limit of slopes of secant lines $= 0$

$$f'(c) = 0$$

Caution: • the converse direction of Fermat's Theorem is not true

Fermat's: (if f has local min/max at c and $f'(c)$ defined) $\Rightarrow f'(c) = 0$

But: $f'(c) = 0$ does not imply f has local min/max at c

e.g. $f(x) = x^3$ $f'(x) = x^2$, $f'(0) = 0$ but
 f does not have local min or max at 0.

• If $f'(c)$ does not exist, f can still have an extreme value at c

e.g. $f(x) = |x|$

$f'(0)$ does not exist, but f has absolute min at 0.

Definition A critical number (aka critical point) of a function f is a number c in the domain of f such that

either $f'(c) = 0$
or $f'(c)$ does not exist.

} Candidate points
for where local min/max
might occur.

Rephrasing of Fermat's Theorem

if f has local min or max at $x=c$,
then c must be a critical point of f

Example Find the critical numbers of $f(x) = x^{3/5} (4-x)$

Find $f'(x)$:

$$\begin{aligned} f'(x) &= \overset{\text{product rule}}{\frac{d}{dx} (x^{3/5}) (4-x) + x^{3/5} \cdot \frac{d}{dx} (4-x)} \\ &= \frac{3}{5} x^{-2/5} (4-x) + x^{3/5} \cdot (-1) \end{aligned}$$

Question: • when is $f'(x)$ undefined? at $x=0$ ← one critical point
because $x^{-2/5} = \frac{1}{x^{2/5}}$ undefined when $x=0$.

• when is $f'(x) = 0$?

$$\frac{3}{5} x^{-2/5} (4-x) + x^{3/5} \cdot (-1) = 0 \quad ?$$

$$\frac{3}{5} \frac{(4-x)}{x^{2/5}} - x^{3/5} = 0 \quad \cdot 5x^{2/5}$$

$$3(4-x) - \underbrace{5x^{2/5} x^{3/5}}_{x^1 = x} = 0$$

$$3(4-x) - 5x = 0 \Rightarrow 12 - 3x - 5x = 0$$

$$12 = 8x$$

$$x = \frac{12}{8} =$$

$$\boxed{\frac{3}{2}}$$

another
critical
point

Final answer: $x=0, x=\frac{3}{2}$

The closed interval method

Suppose a function f is defined over some closed interval $[a, b]$

Goal: Find absolute min and max of f over the interval.

(remember, we are guaranteed absolute min/max exist over $[a, b]$ by extreme value theorem)

- Idea:
- Find all critical points of f in $[a, b]$ (set of candidate points)
 - Separately consider boundary points $x=a, x=b$ (add to set of candidates)
 - evaluate function at all candidate points
 - pick highest value and lowest value

Note: This only works over a closed interval

Example Find the absolute max and min values of the function

$$f(x) = x^3 - 3x^2 + 1 \quad \underline{-\frac{1}{2} \leq x \leq 4}$$

Step 1: Find f' : $f'(x) = \underline{3x^2 - 6x}$

Step 2: Find list of critical points

- $f'(x) = 0$ OR $f'(x)$ ~~undefined~~ always defined in this example

$$3x^2 - 6x = 0 \Rightarrow x(3x - 6) = 0 \Rightarrow \boxed{x=0} \text{ OR } 3x - 6 = 0$$

↑
check those
are in our
interval
→ $\boxed{x=2}$
 $-\frac{1}{2} \leq x \leq 4$ ✓

add endpoints: $-\frac{1}{2}, 4$

list of candidates:

$$\boxed{-\frac{1}{2}, 0, 2, 4}$$

step 3: evaluate f at these points:

x	$f(x) = x^3 - 3x^2 + 1$
$-\frac{1}{2}$	$-\frac{1}{8} - 3 \cdot \frac{1}{4} + 1 = \frac{-1 - 6 + 8}{8} = \frac{1}{8}$
0	$0^3 - 3 \cdot 0^2 + 1 = 1$
2	$2^3 - 3 \cdot 2^2 + 1 = 8 - 12 + 1 = -3$ min
4	$4^3 - 3 \cdot 4^2 + 1 = 64 - 48 + 1 = 17$ max

step 4: identify min and max values in our list:

f attains absolute max value 17 when $x=4$
absolute min value -3 when $x=2$

Example Find the absolute min and max of
 $f(x) = x - 2 \sin(x)$

a) over the interval $[0, 2\pi]$

b) over the interval $[-2\pi, 4\pi]$

Solutions:

a) $f'(x) = 1 - 2 \cos(x)$

• critical pt: $\underline{f'(x)=0}$ OR $f'(x)$ ~~undefined~~
defined in this case

$$1 - 2 \cos(x) = 0 \Rightarrow \cos(x) = \frac{1}{2} \Rightarrow \boxed{x = \frac{\pi}{3}, \frac{5\pi}{3}}$$

- add in boundary points:

Candidate set:

$$\boxed{0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi}$$

- evaluate function:

x	$f(x) = x - 2\sin(x)$
0	$0 - 2 \cdot \sin(0) = 0$
$\pi/3$	$\pi/3 - 2 \sin(\pi/3) = \pi/3 - \sqrt{3} \approx -0.68$ min value
$5\pi/3$	$5\pi/3 - 2 \sin(5\pi/3) = 5\pi/3 + \sqrt{3} \approx 6.97$ max value
2π	$2\pi - 2 \sin(2\pi) = 2\pi \approx 6.28$

b) bigger candidate set ($-2\pi \leq x \leq 4\pi$)

$$x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, -\frac{5\pi}{3}, -\frac{\pi}{3} \quad \text{critical points}$$

add endpoints: $x = -2\pi, 4\pi$

take it from here...

Example: Find the absolute min and max values of

$$f(x) = |2x^2 - 1| + 3x, \quad -2 \leq x \leq 2$$

critical points?

Write f as a piecewise function

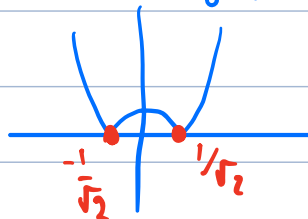
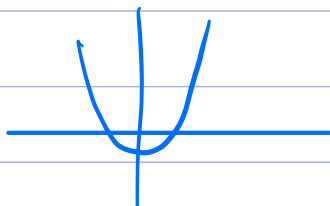
$$|2x^2 - 1| = \begin{cases} 2x^2 - 1 & \text{if } 2x^2 - 1 \geq 0 \\ -(2x^2 - 1) & \text{if } 2x^2 - 1 \leq 0 \end{cases}$$

$$\frac{d}{dx} |2x^2 - 1| = \begin{cases} 4x & \text{if } 2x^2 - 1 > 0 \\ -4x & \text{if } 2x^2 - 1 < 0 \\ \text{undefined} & \text{if } 2x^2 - 1 = 0 \end{cases}$$

$$y = 2x^2 - 1$$

$$y = |2x^2 - 1|$$

Picture



$$f(x) = |2x^2 - 1| + 3x \Rightarrow f'(x) = \frac{d}{dx} |2x^2 - 1| + \frac{d}{dx} (3x)$$

$$f'(x) = \begin{cases} 4x + 3 & \text{if } 2x^2 - 1 > 0 \\ -4x + 3 & \text{if } 2x^2 - 1 < 0 \\ \text{undefined} & \text{if } 2x^2 - 1 = 0 \end{cases}$$

critical points: find when $f' = 0$?

f' undefined?

f' undefined when $2x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{2} \Rightarrow \boxed{x = \pm \frac{1}{\sqrt{2}}}$
critical points

$f' = 0$? Case 1: $4x + 3 = 0$ if $2x^2 - 1 > 0$
 $x = \left(-\frac{3}{4}\right)$ $x^2 = \frac{9}{16} > \frac{1}{2} \checkmark$ $x^2 > \frac{1}{2} \Rightarrow x > \frac{1}{\sqrt{2}}$ or $x < -\frac{1}{\sqrt{2}}$
critical pt

Case 2: $-4x + 3 = 0$ if $2x^2 - 1 < 0$
 $4x = 3$ $x^2 < \frac{1}{2}$
 ~~$x = \frac{3}{4}$~~ $x^2 = \frac{9}{16} > \frac{1}{2} \times$
not a critical point

Candidate set: $\underbrace{-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{3}{4}}_{\text{critical}}, \underbrace{-2, 2}_{\text{boundary}}$

Evaluate function on candidates:

x	$f(x) = 2x^2 - 1 + 3x$
$-\frac{1}{\sqrt{2}}$	$ 2 \cdot \frac{1}{2} - 1 - 3/\sqrt{2} = -3/\sqrt{2}$
$1/\sqrt{2}$	$ 2 \cdot \frac{1}{2} - 1 + 3/\sqrt{2} = 3/\sqrt{2}$
$-3/4$	$ 2 \cdot \frac{9}{16} - 1 - 9/4 = \frac{2}{16} - \frac{9}{4} = \frac{1}{8} - \frac{18}{8} = \boxed{-\frac{17}{8}}$ min value

$$\begin{array}{l|l} -2 & | 2 \cdot 2^2 - 11 - 3 \cdot 2 = 7 - 6 = 1 \\ 2 & | 2 \cdot 2^2 - 11 + 3 \cdot 2 = 7 + 6 = 13 \end{array}$$

max value

$$\frac{17}{8} > \frac{3}{\sqrt{2}}$$